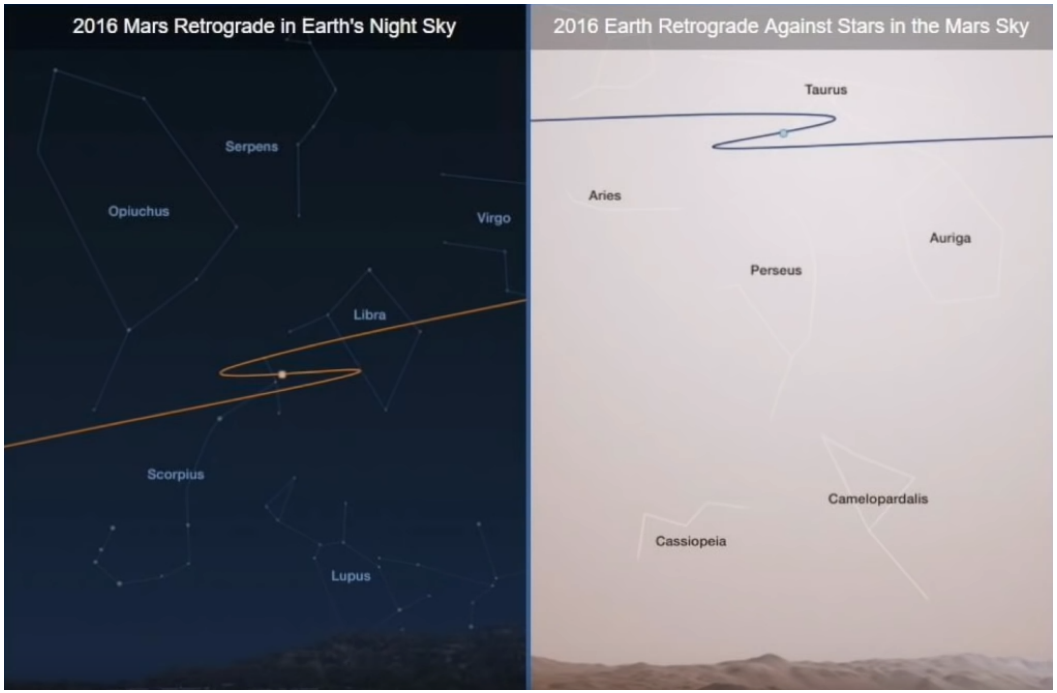
Lecture 9: Linear regression is a geocentric model

- Mars in Earth's sky: `https://player.vimeo.com/video/507033143?h=65b2d62b16`, June 2020-February 2021
- Ubiquity of the normal distribution.

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Retrograde motion of planets



View from Mission Control, Pasadena, CA, USA, Earth

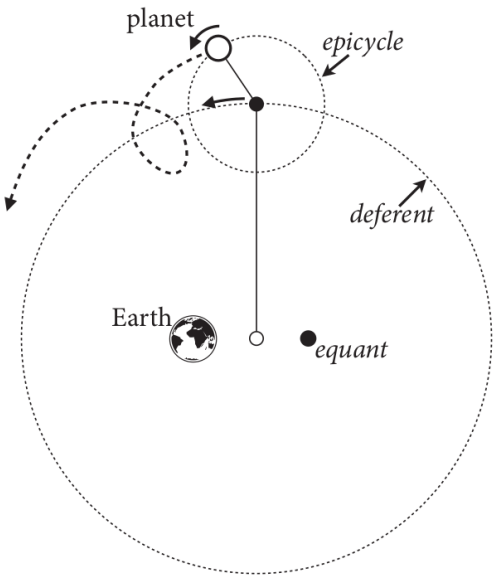
View from Curiosity Rover, Gale Crater, Mars

True explanation: the solar system

- See the retrograde motion of Mars in the solar system explanation (on Stream too).
- But...

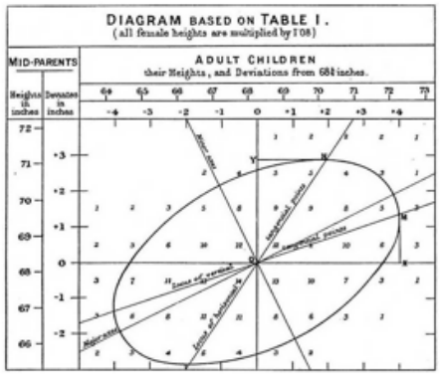
Alternative explanation: the geocentric model

- If you insist on putting Earth in the centre, it works! From the perceived path point of view.
- See the Geocentric model video of Mars' motion around Earth (on Stream too)
- An example of prediction without explanation. Or actually a wrong explanation.

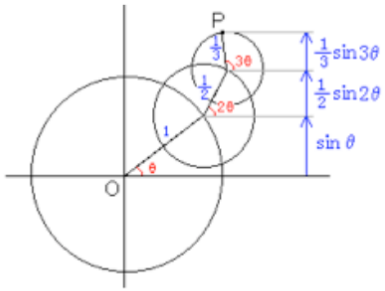


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Linear regression and geocentrism



- descriptively accurate
- mechanistically wrong
- general method of approximation
- taken too seriously



- descriptively accurate
- mechanistically wrong
- general method of approximation
- known to be wrong

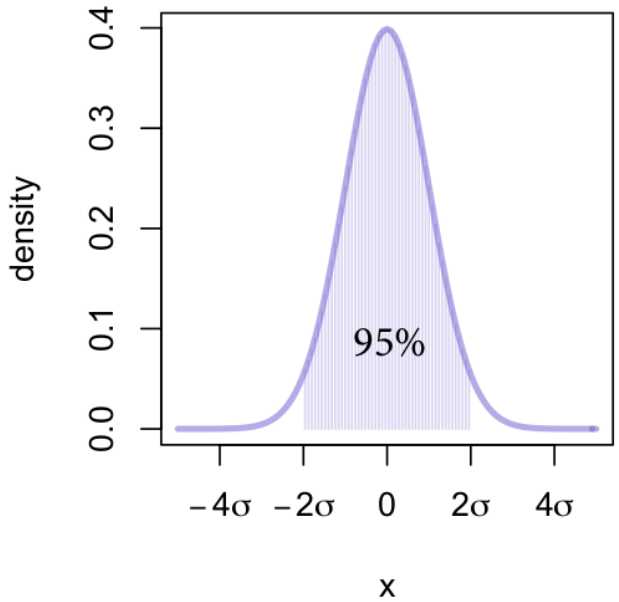
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What linear regression really is

- Simple statistical golems (models)
  - Model of mean and variance of normally (Gaussian) distributed measures. Other distributions exist and we may touch on it towards the end of the course.
  - In the simple form, the mean is decomposed an an additive combination of weighted variables.
  - And the variance is constant.
- Many special cases: ANOVA, ANCOVA, t-test, MANOVA, etc.
- Can also be generative models

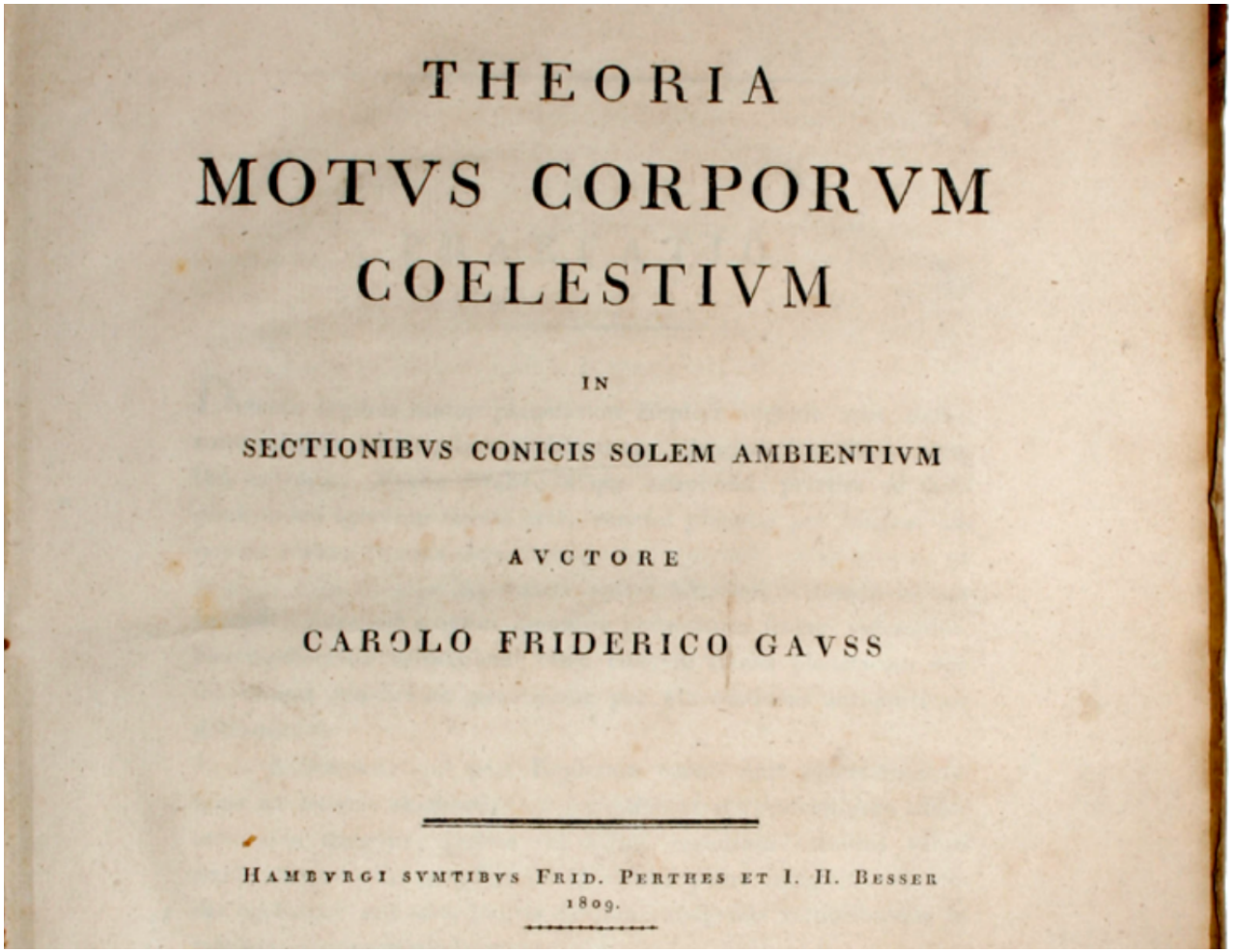
Why normal?

- Why are normal (Gaussian) distributions so common in statistics
  - Easy to calculate with.
  - Common in nature. But not ubiquitous.
  - very conservative assumption.



Bayesian argument for normal error terms

and least-square estimation



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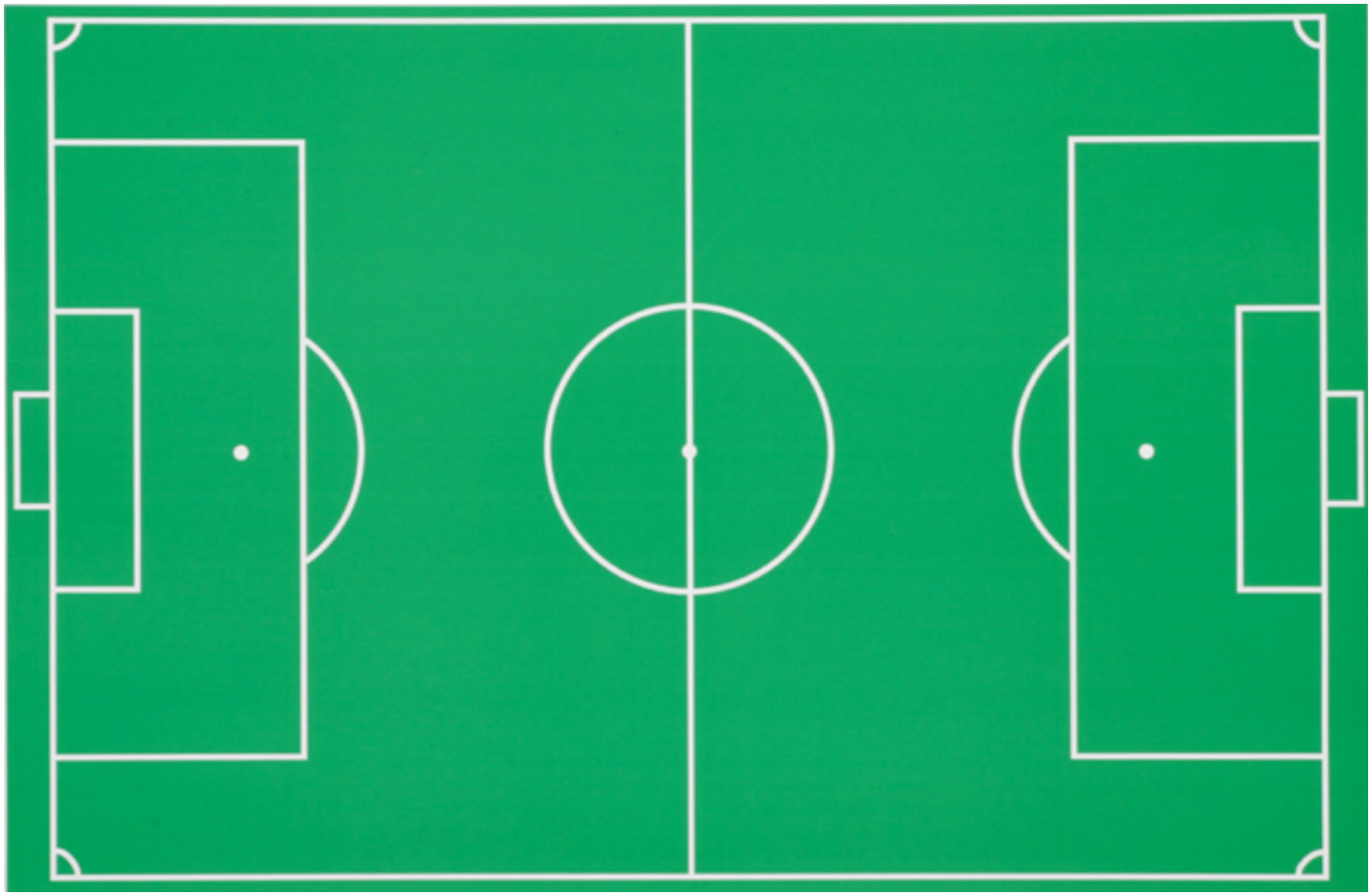
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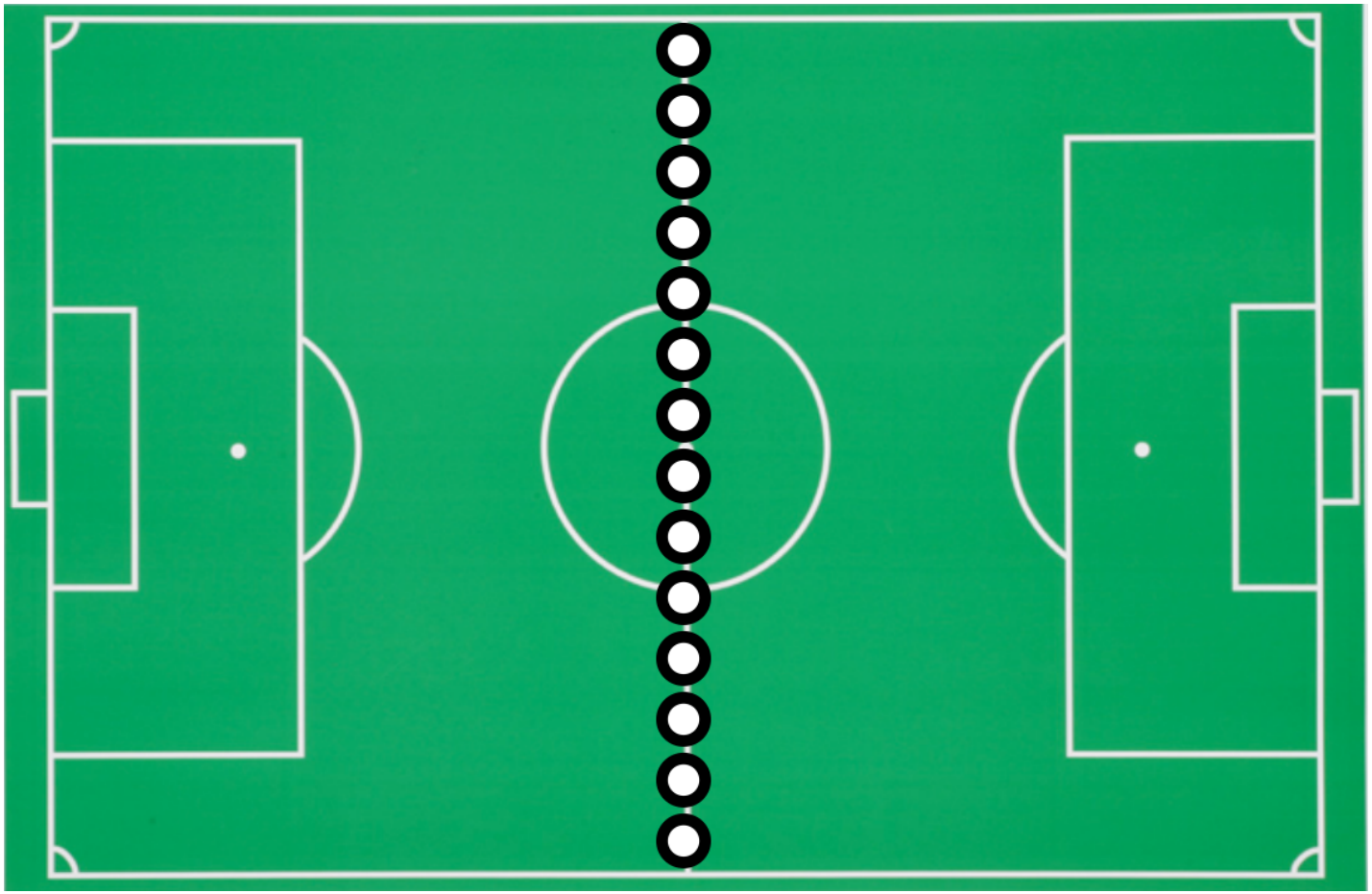
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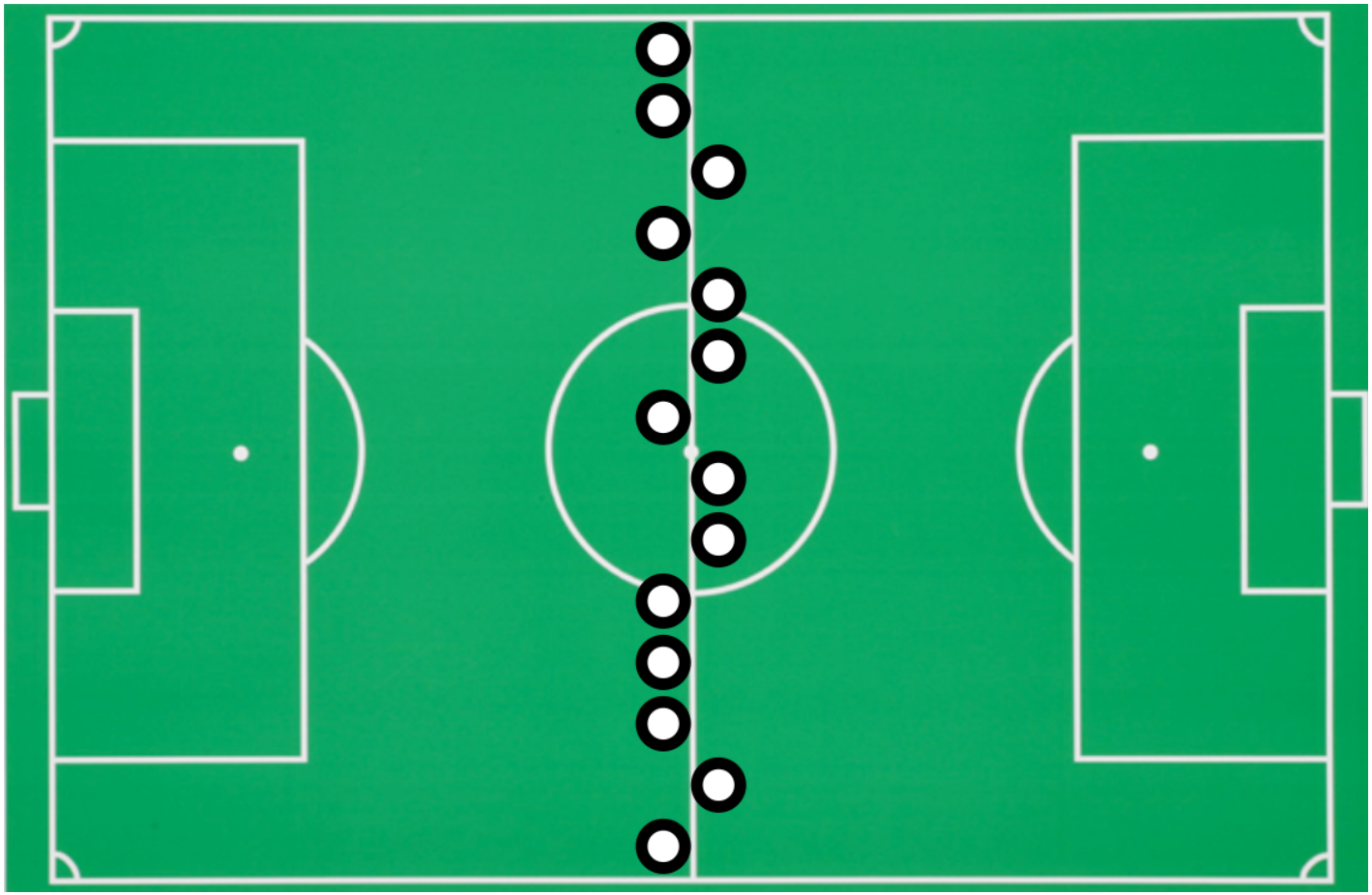
Soccer field with random (Binomial) moves



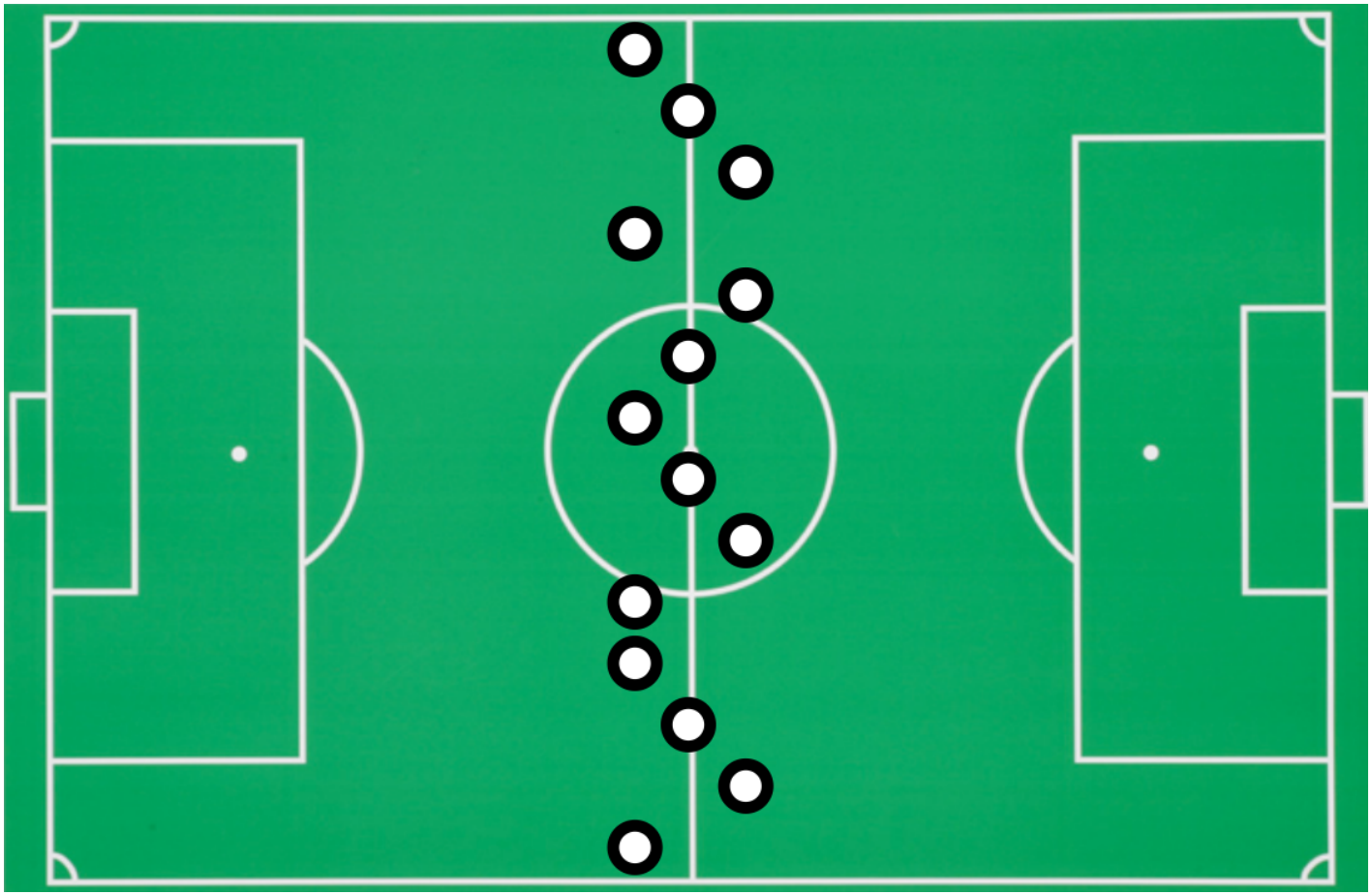
Soccer field with random (Binomial) moves



Soccer field with random (Binomial) moves

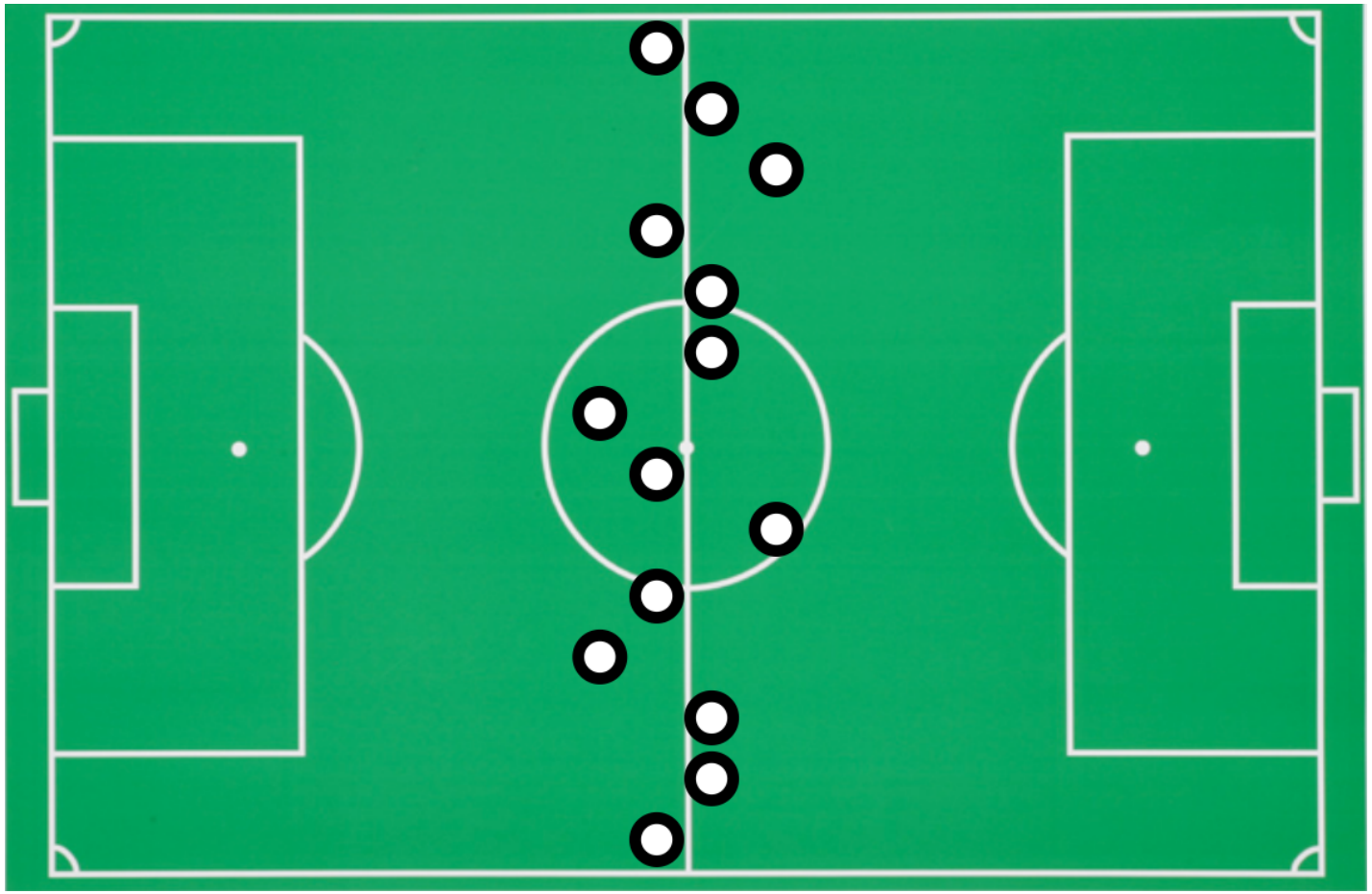


Soccer field with random (Binomial) moves

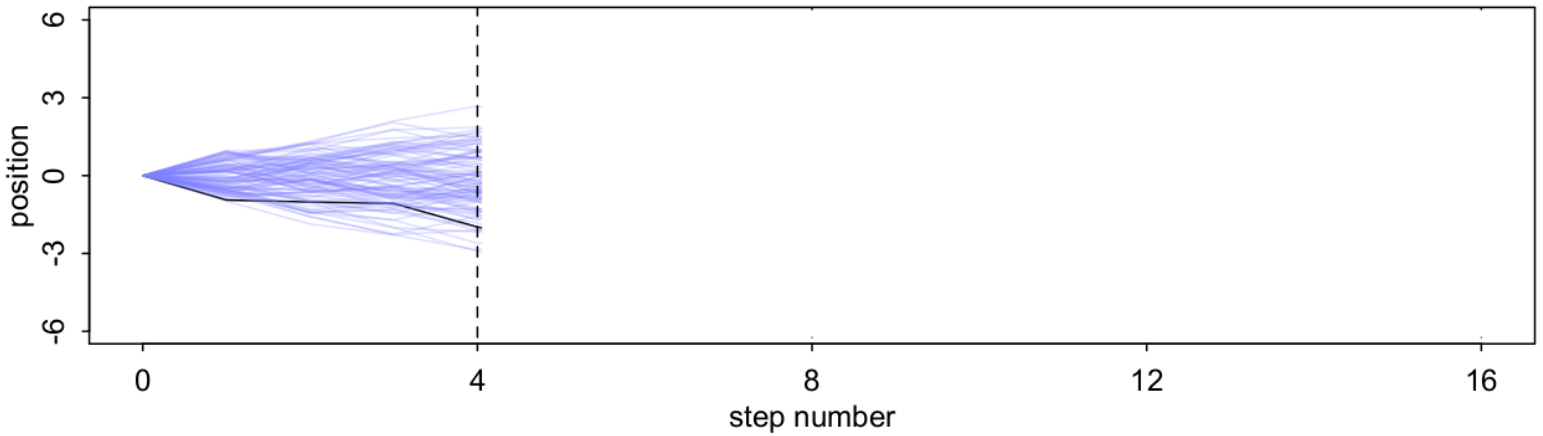


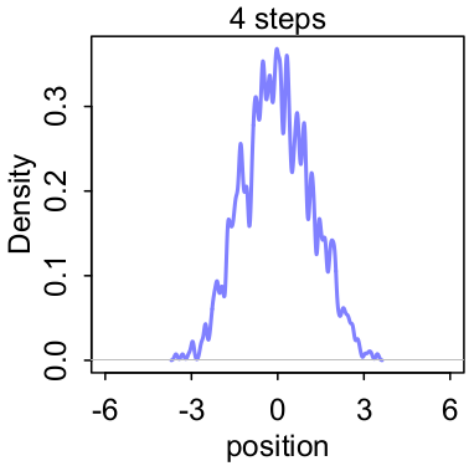
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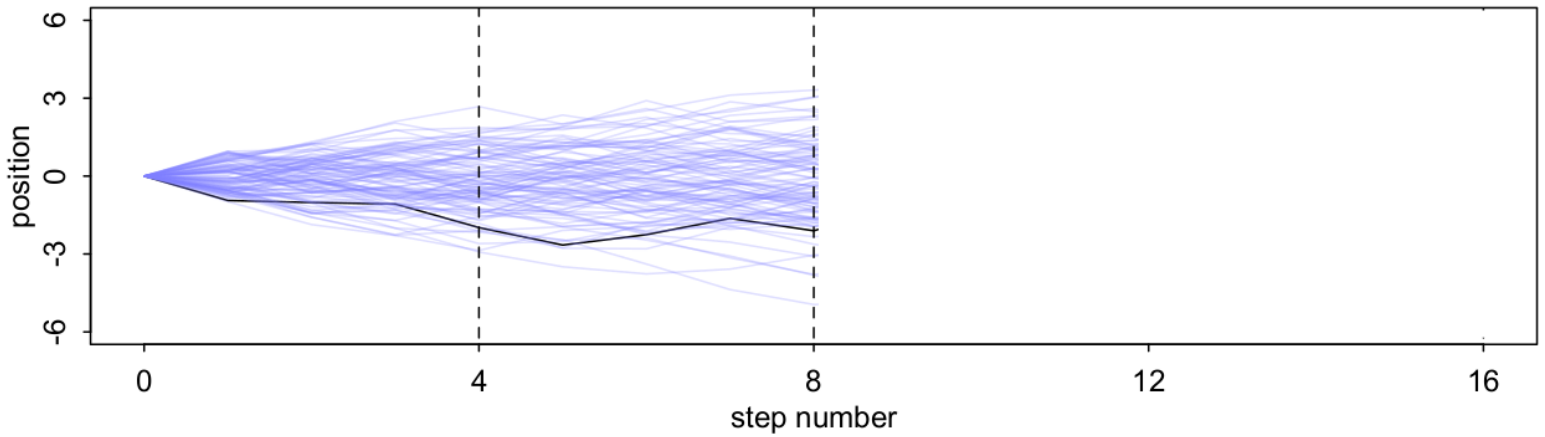


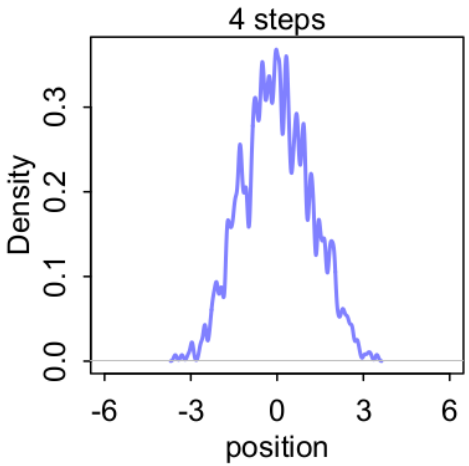
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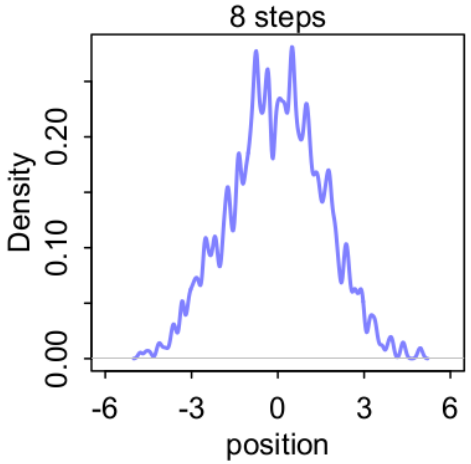




Soccer field with random (Binomial) moves

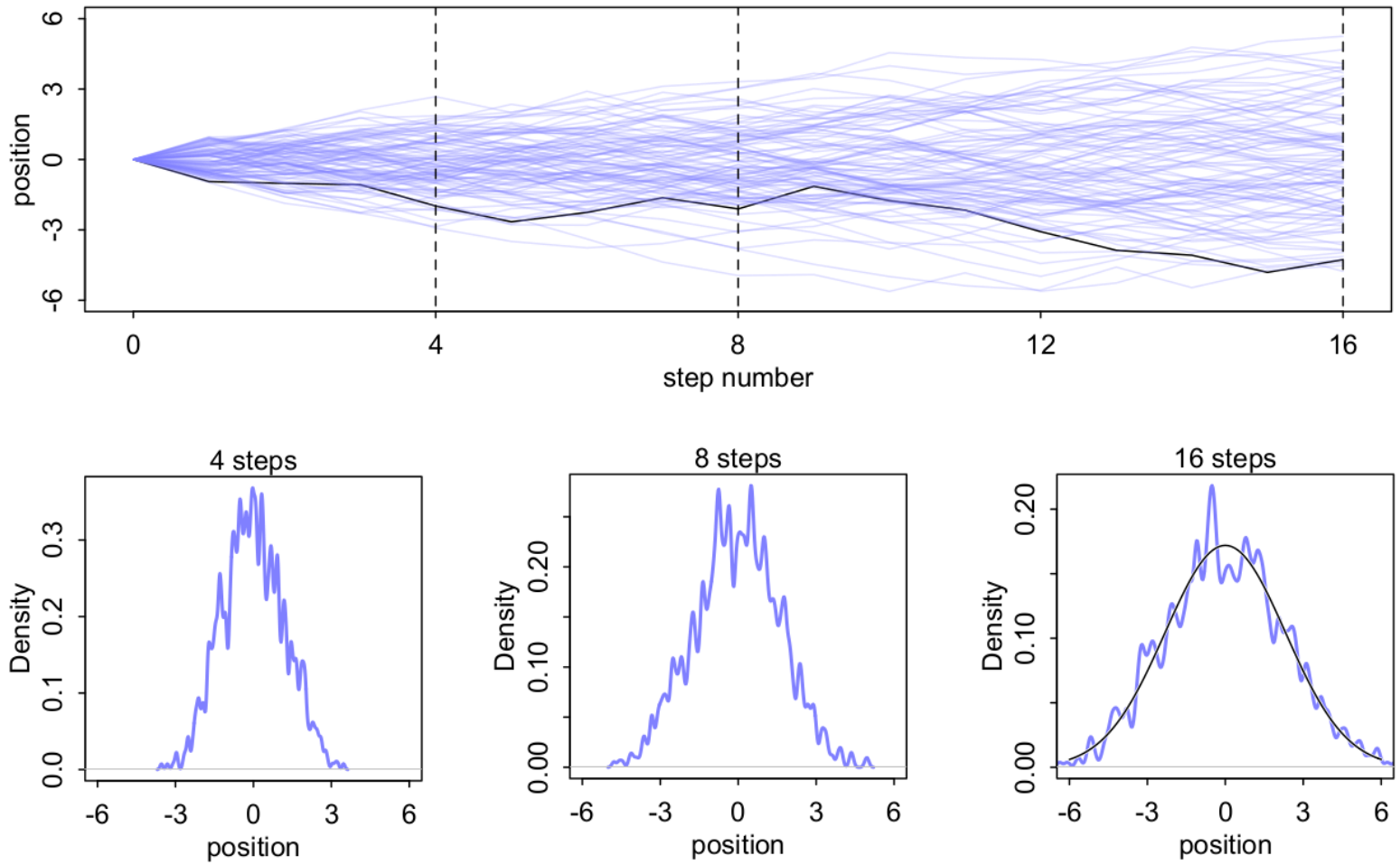






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Soccer field with random (Binomial) moves



Why normal??

- Processes that produce normal distributions:
  - Additions (Central Limit Theorem)
  - Products of small deviations
  - Logarithm of products
- Ontological perspective: dampening with no information left but mean and variance
- Epistemological perspective: mean and variance known only, then least surprising and most conservative (max entropy, nature likes it!) is Gaussian



Francis Galton's 1894 "bean machine" to simulate normal distributions

Central limit theorem

(a simple form)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ_X and variance σ_X^2. Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ to be sample mean. Then for large n, sampling distribution of $\bar{X}$ is approximately $\mathrm{N}(\mu_X, \sigma_X^2/n)$.

So, asymptotically:

- The empirical mean is unbiased
- We know its standard error, $\frac{\sigma}{\sqrt{n}}$

It extends to multiple parameters with a multivariate normal distribution.

Some practice

- Now jump to play with R